



DevFusion: The Developer Hackathon



[RULES HERE](#)

Choose any 1 Problem statement and start doing.

#26ENPRE1: Problem Statement 1

PrepGrid — AI-Powered Interview & Practice Platform

The Challenge: Build a complete placement preparation platform that helps engineering students prepare for technical interviews through AI-driven mock tests, coding practice, and personalised feedback.

Core Features You Must Build:

Practice Module

- Coding question bank with at least 3 difficulty levels (Easy, Medium, Hard)
- In-browser code editor with syntax highlighting and basic run functionality (use Judge0 API — free tier available)
- Topic-wise filtering: Arrays, Strings, Trees, DP, Graphs, SQL, etc.
- Bookmarking questions and tracking solved/unsolved status per user

AI Interview Module

- AI-powered mock interview where the system asks technical questions one by one based on chosen role (Frontend, Backend, Full Stack, DSA)
- User types or speaks their answer — system evaluates it and gives a score + improvement suggestion
- Session history stored so the user can review past interview performances
- Difficulty adapts based on previous answers in the same session

AI Test / Quiz Module

- Generate a timed MCQ test on a chosen topic (e.g. "React Hooks", "DBMS Normalization", "OS Scheduling")
- AI evaluates short-answer questions and gives feedback
- Leaderboard showing top scorers for each topic

User System

- Full authentication (register, login, forgot password with OTP via email)
- User dashboard showing: questions solved, interviews attempted, average score, weak areas
- Streak tracker (days active in a row)

Payments

- Free tier: 5 AI interviews/month, 10 practice questions



DevFusion: The Developer Hackathon

- Pro tier: Unlimited access — integrate Razorpay/Stripe sandbox checkout
- Show a billing history page (even if sandbox)

Admin Panel

- Add/edit/delete questions from the bank
 - View all registered users and their activity
 - Basic analytics: most attempted questions, average scores by topic
-



DevFusion: The Developer Hackathon

#26ENSB2: Problem Statement 2

🎓 SkillBridge — Peer Learning & Doubt Resolution Platform

The Challenge: Build a full-stack platform where students can post doubts, answer questions, follow mentors, attend live doubt sessions, and earn reputation points — like a Stack Overflow built specifically for college students.

Core Features You Must Build:

Doubt Feed

- Post a doubt with rich text (bold, code blocks, images)
- Tag questions by subject, branch, and semester
- Upvote/downvote answers
- Mark one answer as accepted — gives the answerer reputation points
- Filter by: Unanswered, Trending, My Subject, My Branch

AI Doubt Solver

- Before posting, user can "Ask AI First" — AI gives an instant answer
- If unsatisfied, they can post it to the community
- AI explains concepts step by step, not just gives the answer

Mentor System

- Students can apply to become mentors (admin approves)
- Mentors can schedule live doubt sessions with a meeting link (Google Meet / Jitsi)
- Session booking system — students book a 30-min slot from mentor's available times
- Payments: Mentors can set a fee (₹0 to ₹500). Sandbox payment on booking

Practice Tests

- Topic-based MCQ tests (auto-generated via AI for a given subject + topic)
- Timer, auto-submit, instant result with correct answers
- Score history per user per subject

Reputation & Gamification

- Points for: answering doubts (+10), accepted answer (+25), daily login (+2), test completion (+5)
- Badges: First Answer, Helpful Mentor, Streak Master, Subject Expert
- Public profile showing reputation, badges, doubts answered

User System

- Email + Google OAuth login
- Profile: college, branch, semester, subjects, bio, social links
- Notification system: "Your doubt got an answer", "Mentor session in 30 minutes"



DevFusion: The Developer Hackathon

#26ENHL3: Problem Statement 3

HireLoop — End-to-End Campus Recruitment Portal

The Challenge: Build a complete campus placement management system connecting students, college placement cells, and recruiting companies — with AI-powered resume analysis and mock interview features.

Core Features You Must Build:

Three User Roles

- **Student:** Profile, resume upload, job applications, interview preparation
- **Recruiter (Company):** Post jobs, view applicants, shortlist, schedule interviews
- **Placement Cell (Admin):** Manage all companies, approve job posts, view student stats, generate reports

Student Side

- Build and download a resume directly from the platform (select template, fill sections, export as PDF)
- AI Resume Analyser: upload resume → get ATS score, missing keywords, improvement suggestions
- Apply to posted jobs with one click
- Track application status: Applied → Shortlisted → Interview Scheduled → Offer → Rejected
- AI Mock Interview: role-specific questions, answer evaluation, session summary

Recruiter Side

- Post job openings: role, eligibility (CGPA, branch, batch), deadline, salary range
- View all applicants with filters (CGPA, branch, skills)
- Shortlist applicants and schedule interview slots
- Send automated email notifications on status changes

Placement Cell Side

- Approve or reject company registrations
- Dashboard: total students placed, companies registered, packages offered
- Generate placement report by branch, batch, or company (downloadable)
- Announcement board: post notices visible to all students

Payments

- Companies pay a listing fee to post jobs (sandbox checkout)
- Students can upgrade to Premium: AI interview unlimited + priority visibility to recruiters

AI Features



DevFusion: The Developer Hackathon

- Resume scoring against a job description (match percentage)
 - Auto-generate cover letter draft from resume + job description
 - Smart job recommendations based on student skills and past applications
-



DevFusion: The Developer Hackathon

#26ENCF4: Problem Statement 4

CourseForge — Student-Built Course Marketplace

The Challenge: Build a full-featured online learning platform where students and educators can create, sell, and learn from courses — with AI-powered learning assistance and progress tracking.

Core Features You Must Build:

Course Creation (Instructor Side)

- Create a course: title, description, thumbnail, category, price (free or paid)
- Upload video lessons (use Cloudinary free tier or similar for storage)
- Add text-based lessons, quizzes, and assignments per module
- Set prerequisites and publish/unpublish toggle
- Earnings dashboard: total revenue, enrollments, course ratings

Learning Experience (Student Side)

- Browse courses by category, rating, price (free/paid filter)
- Video player with progress tracking (resume from where you left off)
- Notes panel alongside video (student can write timestamped notes)
- Download certificate on course completion (generate PDF certificate)
- Rate and review courses

AI Learning Assistant

- "Ask AI" button on any lesson: student types a question about the current topic → AI explains
- AI generates a short quiz based on the lesson content the student just watched
- Adaptive learning path: AI suggests what to learn next based on current progress and goals

AI Mock Test

- Student picks a topic or course → AI generates a 20-question timed test
- Detailed result: topic-wise breakdown, questions with explanations, improvement tips
- Leaderboard per course among enrolled students

Payments

- Sandbox payment gateway for paid course enrollment
- Instructor payout simulation (show earnings, withdrawal history)
- Coupon/discount code system

Community

- Q&A section per course: students post questions, instructors and peers answer



DevFusion: The Developer Hackathon

- Upvote helpful answers
 - Instructor announcements visible to enrolled students
-



DevFusion: The Developer Hackathon

#26ENMT5: Problem Statement 5

MediTrack — Smart Healthcare Management System

The Challenge: Build a complete healthcare web application connecting patients, doctors, and hospital admins — with appointment booking, health record management, AI symptom analysis, and telemedicine features.

Core Features You Must Build:

Three User Roles

- **Patient:** Book appointments, view records, AI health check, video consultation
- **Doctor:** Manage schedule, view patients, write prescriptions, upload reports
- **Admin:** Manage doctors, view hospital analytics, approve registrations

Patient Side

- Search doctors by specialisation, location, rating, availability
- Book appointment: choose date + time slot, select in-person or video call
- AI Symptom Checker: patient enters symptoms → AI gives possible conditions + urgency level (not a diagnosis — clearly labelled as informational)
- View past prescriptions, lab reports, and appointment history
- Medication reminder setup (shown in dashboard, no push notification needed)

Doctor Side

- Availability calendar: set available days and time slots
- Patient queue: today's appointments with status (Waiting, In Progress, Done)
- Write digital prescription: medicine name, dosage, duration, notes
- Upload lab reports and link to patient record
- Video consultation room (integrate Jitsi Meet embed — free and open source)

Admin Side

- Approve or reject doctor registrations (verify credentials manually)
- Dashboard: total patients registered, appointments today, doctors active
- Manage specialisations and hospital departments

AI Features

- Symptom checker with urgency flagging
- AI-generated health tips based on user's appointment and medical history
- Smart doctor recommendation: based on symptoms entered, suggest top 3 specialists

Payments

- Consultation fee payment (sandbox) before confirming appointment
- Billing history for patients



DevFusion: The Developer Hackathon

- Doctor earnings dashboard (simulated)
-



DevFusion: The Developer Hackathon

#26ENVH6: Problem Statement 6

VendorHub — Hyperlocal Multi-Vendor E-Commerce Platform

The Challenge: Build a multi-vendor marketplace where local sellers can list products, manage orders, and grow their business — while buyers get a smooth, modern shopping experience with AI-powered recommendations.

Core Features You Must Build:

Three User Roles

- **Buyer:** Browse, search, cart, checkout, order tracking
- **Seller/Vendor:** Product listing, inventory, orders, earnings
- **Admin:** Platform management, vendor approval, reports

Buyer Side

- Product search with filters: category, price range, rating, location of vendor
- Product detail page: images, description, reviews, seller info
- Cart and wishlist functionality
- Checkout with address selection and sandbox payment
- Order tracking with status updates: Placed → Confirmed → Shipped → Delivered
- Review and rating after delivery

Seller Side

- Register as a vendor (admin approval required)
- Add/edit/delete products: name, images (multiple), description, price, stock, category
- Order management: view incoming orders, update status (Confirmed → Shipped)
- Earnings dashboard: total revenue, orders this week, top-selling products
- Inventory alerts: low stock warning

Admin Side

- Approve or reject vendor registrations
- Platform analytics: total sales, top vendors, top categories
- Manage categories and subcategories
- Handle refund requests (mark as approved/rejected)
- Commission settings (show vendor earnings after platform commission is deducted)

AI Features

- Product recommendations based on browsing history and past orders
- AI-powered search: fuzzy matching and synonym understanding ("laptop bag" also shows "notebook carry case")



DevFusion: The Developer Hackathon

- Smart price suggestion for sellers: based on similar products listed on the platform

Payments

- Razorpay/Stripe sandbox checkout with order summary
 - Refund flow (simulated)
 - Vendor payout history (simulated)
-



DevFusion: The Developer Hackathon

#26ENNF7: Problem Statement 7

NestFinder — Smart Real Estate Discovery Platform

The Challenge: Build a full-stack real estate platform where property owners list homes, buyers and renters discover properties, and AI helps match users to the right property based on preferences and budget.

Core Features You Must Build:

Listings System

- List a property: type (rent/buy), category (flat/house/PG/hostel/commercial), location, price, amenities, photos
- Map integration using Leaflet.js (free, open source) to show property location
- Advanced filters: city, locality, price range, BHK type, furnishing, amenities (WiFi, parking, gym)
- Save/favourite properties

AI-Powered Matching

- Onboarding quiz: budget, preferred location, purpose (rent/buy), required amenities, lifestyle (pet-friendly, family, bachelor)
- AI suggests top 5 matching properties with explanation of why each was recommended
- "Find Similar" button on any listing — shows comparable properties

AI Chatbot

- Property-specific chatbot: user can ask "Is this near a metro?" or "What's the society maintenance?" — AI answers based on listing data
- General assistant: "Show me 2BHK flats under ₹15,000/month in Lucknow"

User Roles

- **Owner/Agent:** List properties, manage inquiries, mark as rented/sold
- **Buyer/Renter:** Search, save, book site visits, contact owner
- **Admin:** Verify listings, manage reported listings, view platform stats

Booking & Payments

- Schedule a site visit: pick a date and time, owner gets notified
- Token amount payment for booking a property (sandbox checkout)
- Rental agreement download (pre-filled PDF template based on listing data)

Community & Trust

- Verified listing badge (admin reviews and approves)
- Owner ratings by past tenants



DevFusion: The Developer Hackathon

- Neighbourhood insights: nearby schools, hospitals, transport (use static data or OpenStreetMap)
-



DevFusion: The Developer Hackathon

#26ENES8: Problem Statement 8

🎯 EventSphere — End-to-End Event Management & Ticketing Platform

The Challenge: Build a complete event management platform where organisers can create events, attendees can register and buy tickets, and AI helps with event discovery and personalisation.

Core Features You Must Build:

Event Creation (Organiser Side)

- Create event: name, date/time, venue (physical or online), category, description, banner image
- Ticket types: Free, Paid (single price or multiple tiers like General, VIP, Early Bird)
- Set capacity per ticket type
- Discount codes and early bird pricing with expiry dates
- Organiser dashboard: registrations, revenue, check-in stats, attendee list (downloadable CSV)
- QR code generation for each ticket — attendees show this at entry

Attendee Side

- Browse events: filter by category, date, city, price (free/paid)
- Event detail page: agenda, speakers, venue map, FAQ
- Register or buy tickets (sandbox payment)
- Receive ticket confirmation with QR code (shown on dashboard, downloadable)
- Wishlist events and get reminder notifications (shown in dashboard)

Check-in System

- Organiser can open a check-in panel on event day
- Scan QR code (simulate with a manual entry field) to mark attendee as checked in
- Live count: total registered vs checked in

AI Features

- Event recommendation: based on past events attended and categories saved
- AI-generated event description: organiser enters bullet points → AI drafts a polished description (limited use case — this is acceptable as it's a tool feature, not building the app)
- Smart schedule builder: for multi-session events, AI suggests optimal session ordering based on speaker availability and audience flow

Payments

- Multi-ticket checkout (buy 2 VIP + 1 General in one order)
- Razorpay/Stripe sandbox with order summary



DevFusion: The Developer Hackathon

- Refund request flow (organiser approves or rejects)
- Organiser payout simulation

Community

- Post-event feedback form (auto-sent after event ends)
- Public event reviews and ratings
- Attendee networking: opt-in to share your LinkedIn with other attendees of the same event